1 Basic Information

Term	Spring 2026
Course Title	電腦硬體與程式語言在行為科學實驗與大數據分析之應用 The Applications of Computer Hardware and Programming Languages in Behavioral Experiments and Big-Data
Course Code	NS5116
Classroom	S5-607-1
Credit Hours	3.0
Class Schedule	Thursdays : 9:00 AM -12:00 PM
Course Teacher	Erik Chang 張智宏
Teacher Email	audachang@gmail.com
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Office and Location	Building: Science 5, Teacher Room: 601
Teacher Office Hours	by appointment
Online Course Link	https://ncueeclass.ncu.edu.tw/course/32506

2 Weekly Schedule

Week	Date	Topic
1	2026/02/26	Environment setup, variables, types, operators
2	2026/03/05	if/else, for/while loops, functions, scope, debugging
3	2026/03/12	Lists, dicts, tuples, sets; CSV & text file I/O
4	2026/03/19	PsychoPy install, Window, visual stimuli, clocks, input
5	2026/03/26	PsychoPy Experiment Design
6	2026/04/02	NumPy arrays, indexing, vectorized operations
7	2026/04/09	Line/scatter/bar/histogram plots, subplots, annotations
8	2026/04/16	*Midterm: Deploy cognitive experiment to Pavlovia*
9	2026/04/23	Vibe/agentic coding concepts; Claude Code CLI setup; prompting
10	2026/04/30	Multi-step task delegation, git commits, GitHub Actions
11	2026/05/07	Streamlit app; layout, widgets, charts; deploy to Streamlit Cloud
12	2026/05/14	Taiwan open data portal, API calls, data cleaning
13	2026/05/21	Interactive charts, narrative data storytelling
14	2026/05/28	Anthropic SDK; add AI features; responsible AI
15	2026/06/04	Final project workshop; peer code review; polish UI/UX
16	2026/06/18	*Final Presentation: Present online app*

3 Course Resources

- **GitHub Repository**: https://github.com/audachang/ai-programming-psychology
- **Google Classroom**: https://classroom.google.com/c/ODQwMzEwNDQ3NjI5?cjc=hfwtvn66

4 Grading Policy

Component	Description
Assignments (40%)	Four practical coding assignments: 1. PsychoPy Experiment Implementation 2. Statistical Analysis & Visualization 3. Machine Learning Model Training (Regression/Classification) 4. Advanced AI Pipeline (GPU/Deep Learning/LLM) **All work must be synced to individual GitHub repositories.**
Midterm Project (20%)	**Experimental Demo**: Demonstration of a custom-built psychological task using Python, focusing on creativity and technical precision.
Final Capstone (30%)	**End-to-End AI Project**: A complete pipeline from data source to AI prediction. Presented in a professional Poster Session format.
Participation (10%)	Code review contributions and active participation in class discussions.

5 Weekly Content Details

Module 1: Experimental Programming & Data Science

Focus: *From Hypothesis to Data—Mastering the Python ecosystem to build robust behavioral experiments, ensure reproducibility, and visualize scientific results.*

Week 1: Orientation, Python Environment Setup & Basic Libraries

- **Date**: 2026/02/26

- **Topic**: **Setting the Stage for Scientific Computing**
- **Course Content**:
  - **The Stack**: VS Code, Anaconda/Miniconda, and managing Virtual Environments (`conda/venv`).
  - **Python Refresher**: Mutable vs. Immutable types, List Comprehensions, and Modular design.
  - **Scientific Libs**: Introduction to **NumPy** for high-performance array manipulation (essential for coordinate systems, timing, and randomization).
- **Teaching Plan**:
  - **Lecture**: The ”Reproducibility Crisis” and how code provides the solution.
  - **Lab Activity (Install-Fest)**: Ensure every student has a working local environment.
  - **Homework**: ”Hello Data”—Write a script that generates a synthetic dataset of reaction times using NumPy.

Week 2: PsychoPy Coder & Stimulus Presentation

- **Date**: 2026/03/05
- **Topic**: **Drawing to the Screen (Beyond the GUI)**
- **Course Content**:
  - **The Window**: `visual.Window`, units (`norm`, `pix`, `deg`), and frame buffering (`win.flip()`).
  - **Stimuli**: Programmatically creating `TextStim`, `ImageStim`, and `GratingStim` (Gabor patches).
  - **Timing**: Understanding refresh rates (Hz), frame-based timing, and precision.
- **Teaching Plan**:
  - **Live Coding**: Build a script from scratch that displays a fixation cross followed by a drifting Gabor patch.
  - **Lab**: Modify the script to change the orientation of the grating dynamically based on a frame counter.

Week 3: Interaction & The Event Loop

- **Date**: 2026/03/12
- **Topic**: **The Game Loop of Research**
- **Course Content**:

- **Input**: Polling (`defaultKeyboard.getKeys()`) vs. Blocking (`event.waitKeys()`).
- **Architecture**: Implementing the Trial → Block → Experiment hierarchy.
- **Data Logging**: Using `pandas` or PsychoPy’s `ExperimentHandler` to save trial-by-trial data to CSV.

- **Teaching Plan**:
  - **Lab Activity (The Stroop Task)**: Build a complete interactive task where participants identify font colors of conflicting words.
  - **Crucial Step**: Students must run the task on themselves to generate real data for Week 5.

Week 4: PsychoPy Builder, Online Paradigms & Adaptive Design

- **Date**: 2026/03/19
- **Topic**: **Rapid Prototyping & Web Deployment**
- **Course Content**:
  - **Builder GUI**: Routines, Loops, and injecting Custom Code components.
  - **Pavlovia**: Converting Python experiments to PsychoJS for online data collection.
  - **Adaptive Design**: Introduction to **Staircase procedures** (e.g., Up/Down method) for finding sensory thresholds.
- **Teaching Plan**:
  - **Demo**: Recreate the Week 3 manual code using the Builder GUI in 15 minutes.
  - **Workshop**: Push a simple experiment to Pavlovia and collect data from a peer.

Week 5: PsychoPy Experiment Design

- **Date**: 2026/03/26
- **Topic**: **From Raw Logs to Insights**
- **Course Content**:
  - **Data Cleaning**: Using **Pandas** to filter outliers, group data, and handle missing values.
  - **Visualization**: **Seaborn** and **Matplotlib** for Violin plots, Scatter plots, and Error bars.
  - **Stats**: T-tests and correlations using `scipy.stats`.

- **Teaching Plan**:
  - **Data Dive**: Analyze the **Stroop Task data** generated in Week 3.
  - **Deliverable**: A script that produces a publication-quality figure showing the "Stroop Effect" (RT difference).

Week 6: NumPy arrays, indexing, vectorized operations

- **Date**: 2026/04/02
- **Topic**: **Deconstructing Classic Paradigms**
- **Course Content**:
  - **Paradigm 1**: Posner Cueing Task (Attentional Orienting).
  - **Paradigm 2**: n-Back Task (Working Memory).
  - **Logic**: Blocked vs. Interleaved designs; Counterbalancing conditions programmatically.
- **Teaching Plan**:
  - **Group Analysis**: Students break down the logic of these tasks and write the "pseudocode" structure before implementation.

Week 7: Line/scatter/bar/histogram plots, subplots, annotations

- **Date**: 2026/04/09
- **Topic**: **Modern Coding Workflows**
- **Course Content**:
  - **AI Tools**: Prompt Engineering for GitHub Copilot/ChatGPT (e.g., "Explain this bug," "Refactor for speed").
  - **Clean Code**: Modular functions, docstrings, and PEP8 standards.
  - **Version Control**: Basics of **Git** (`init`, `commit`, `push`) and GitHub.
- **Teaching Plan**:
  - **Hackathon ("Refactor Day")**: Students are given a messy, broken script and must use AI tools to fix, comment, and optimize it.

Week 8: Midterm Project Presentation

- **Date**: 2026/04/16
- **Activity**: **Seminar & Demo**

- **Requirements**:
  1. A fully functional Python/PsychoPy experiment.
  2. Automatic data logging.
  3. A visualization of pilot data.
  4. A clean GitHub repository link.

Module 2: Machine Learning & AI Applications

Focus: *Transitioning from analyzing past data to predicting future outcomes using Machine Learning, Deep Learning, and LLMs.*

Week 9: Machine Learning Foundations

- **Date**: 2026/04/23
- **Topic**: **Concepts & Pipelines**
- **Course Content**:
  - **Concepts**: Supervised vs. Unsupervised learning; Features (X) and Labels (y).
  - **The Pipeline**: Data preprocessing (Scaling/Normalization), One-Hot Encoding, and Train/Test splits.
  - **Library**: Introduction to `scikit-learn`.
- **Teaching Plan**:
  - **Lab**: Preprocessing a behavioral dataset to prepare it for ML (e.g., converting subject IDs to categorical codes).

Week 10: Basic ML Algorithms: Regression & Classification

- **Date**: 2026/04/30
- **Topic**: **Predictive Modeling**
- **Course Content**:
  - **Regression**: Linear Regression (Predicting continuous variables like RT).
  - **Classification**: Logistic Regression and SVM (Predicting binary outcomes like Error/Correct).
  - **Evaluation**: Accuracy, Precision, Recall, Confusion Matrix.
- **Teaching Plan**:

- **Challenge**: Train a model to predict if a subject will make an error on the *next* trial based on their RT in the *previous* trial.

Week 11: Advanced ML Algorithms

- **Date**: 2026/05/07
- **Topic**: **Complexity & Unsupervised Learning**
- **Course Content**:
  - **Ensembles**: Random Forests and Gradient Boosting (XGBoost).
  - **Dimensionality Reduction**: **PCA** (Principal Component Analysis) for high-dimensional data (e.g., surveys/pixels).
  - **Clustering**: K-Means.
- **Teaching Plan**:
  - **Lab**: Use PCA to visualize a high-dimensional dataset in 2D space, then apply K-Means to identify clusters of participants.

Week 12: GPU Acceleration Tools

- **Date**: 2026/05/14
- **Topic**: **High-Performance Computing**
- **Course Content**:
  - **Hardware**: CPU vs. GPU architecture differences.
  - **Tools**: Google Colab (Cloud GPUs), **CuPy** (NumPy on GPU), and **PyTorch Tensors**.
- **Teaching Plan**:
  - **Benchmark Lab**: Write a script that performs massive matrix multiplication. Measure and compare execution time on CPU (NumPy) vs. GPU (PyTorch).

Week 13: Deep Learning

- **Date**: 2026/05/21
- **Topic**: **Neural Networks**
- **Course Content**:
  - **Basics**: Neurons, Layers, Weights, Biases, Activation Functions (ReLU).
  - **Training**: Backpropagation and Optimizers (Adam).

  - **Implementation**: Building a Multi-Layer Perceptron (MLP) in PyTorch.
- **Teaching Plan**:
  - **Lab**: "Hello World" of Deep Learning—Training a network to classify handwritten digits (MNIST) or behavioral patterns.

Week 14: Large Language Model (LLM)

- **Date**: 2026/05/28
- **Topic**: **NLP in Research**
- **Course Content**:
  - **Theory**: Transformer Architecture overview (Attention mechanism).
  - **Tools**: Hugging Face `transformers` library.
  - **Application**: Using LLMs to generate stimuli text or analyze qualitative survey responses.
- **Teaching Plan**:
  - **Workshop**: Build a pipeline that uses a pre-trained BERT model to perform Sentiment Analysis on a text dataset.

Week 15: ML & AI Capstone Studio

- **Date**: 2026/06/04
- **Activity**: **Guided Development / Hackathon**
- **Details**:
  - Students work on their final projects in class.
  - Instructor acts as a technical consultant (debugging code, refining models).
  - **Peer Review**: Students pair up to code-review their analysis pipelines.

Week 16: Final Poster Presentation

- **Date**: 2026/06/11
- **Activity**: **Symposium**
- **Deliverables**:
  1. **Scientific Poster**: Introduction, Methods (Code/Algo), Results (Vis/Stats), Conclusion.
  2. **Code Repository**: A public GitHub link allowing others to reproduce the results.